General Bite #6: Power Supplies and Schematic Symbols

Half-wave vs full-wave rectifiers, filters, bleeder resistors, switchmode vs linear supplies, and recognizing parts on Figure G7-1.

Group G7A · 13 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

What A Linear Power Supply Does

Every radio runs on DC. The wall gives AC. A power supply turns AC into clean DC in three stages:

Stage 1: Rectifier – diodes that pass current in one direction, blocking the other half of the AC cycle.
Stage 2: Filter – **capacitors and inductors** (G7A02) smooth the rectifier's pulsing output into steady DC.
Stage 3: Regulator – holds the DC voltage constant under load and line changes.
An unfiltered rectifier output is a series of DC pulses, not steady DC. The filter network is what makes it usable.

Rectifier Types

Type	Behavior	Q
Half-wave	Uses **one diode** (G7A04). Converts **180° of each AC cycle** to DC (G7A05) – the other half is blocked. Output ripple = AC frequency (60 Hz from 60 Hz mains).	G7A04, G7A05
Full-wave (center-tap)	Uses **two diodes and a center-tapped transformer** (G7A03). Converts **360° of each AC cycle** (G7A06). Output is **DC pulses at twice the AC frequency** (G7A07) – 120 Hz from 60 Hz mains.	G7A03, G7A06, G7A07
Full-wave bridge	Four diodes, no center tap. Also converts 360°.	—

Bleeder Resistor

Function of a power-supply bleeder resistor: It discharges the filter capacitors when power is removed (G7A01). Without one, large filter caps can hold a lethal voltage for a long time after the supply is unplugged.

Switchmode vs Linear Supply

Characteristic of a switchmode supply vs a linear supply: High-frequency operation allows the use of smaller components (G7A08).
A linear supply uses a 60 Hz mains transformer (large, heavy) and dissipates the difference between rectified voltage and regulated voltage as heat. A switchmode supply chops the input at tens or hundreds of kHz, so the transformer and filter parts shrink dramatically, and the pass element is either fully on or fully off (very efficient).

Figure G7-1: Schematic Symbol Recognition

The pool's only General-class figure is **Figure G7-1**, a chart of numbered schematic symbols. Five exam questions ask you to identify a specific symbol by number. The figure is at the NCVEC site (`ncvec.org/downloads/G7-1.pdf`).

Memorize these five symbol-number assignments:

- **Symbol 1** = **Field-effect transistor (FET)**. (G7A09) Three terminals labeled G, D, S. Gate lead is separated from the channel (no direct connection).
- **Symbol 2** = **NPN bipolar junction transistor**. (G7A11) Circle with B, C, E and arrow on emitter pointing out.
- **Symbol 5** = **Zener diode**. (G7A10) Diode triangle with bent/stepped cathode bar.
- **Symbol 6** = **Solid-core transformer**. (G7A12) Two coils with solid lines between them (the core).
- **Symbol 7** = **Tapped inductor**. (G7A13) Coil with a tap line at one point.

Note: Figure G7-1 has not been embedded in this handout. Print the official figure from the NCVEC site before your study session, or pull it up on screen during the meeting. The five identifications above are all the pool tests from this figure.

Mental Cheat Sheet

- Bleeder R = discharges filter caps when off.
- Filter network = **capacitors and inductors**.
- Half-wave: **1 diode**, **180°** of cycle.
- Full-wave center-tap: **2 diodes + center-tapped transformer**, **360°**.
- Full-wave output ripple = **2x AC frequency** (120 Hz from 60 Hz).
- Switchmode advantage = **smaller components** via high-frequency operation.
- Figure G7-1:
  - Symbol 1 = FET
  - Symbol 2 = NPN
  - Symbol 5 = Zener diode
  - Symbol 6 = solid-core transformer
  - Symbol 7 = tapped inductor

Common Traps

- Bleeder R discharges caps – it is **not** a fuse, not a ground-loop killer.
- Filter components are L and C; diodes are the rectifier, not the filter.
- Half-wave converts **180°**, full-wave converts **360°**. Don't swap.
- Full-wave output ripple is **twice** the AC frequency, not the same as it.
- Switchmode supplies are smaller because of **high-frequency operation**, not "fewer components" or "more stable."
- Figure G7-1: the **NPN** is symbol **2** (not 1). The **FET** is symbol **1** (not 2). Easy swap.

- The **Zener** is symbol **5**, the **solid-core transformer** is symbol **6**, the **tapped inductor** is symbol **7**. Memorize the order.

Takeaway: AC comes in, diodes rectify (half-wave = 1 diode/180°, full-wave = 2 diodes/360°), L+C filters smooth, regulator holds steady. Bleeder R is the safety drain for filter caps. Switchmode supplies trade complexity for size by chopping at high frequency. Figure G7-1: memorize symbols 1 (FET), 2 (NPN), 5 (Zener), 6 (transformer), 7 (tapped inductor).

Bite 6 Question Bank (G7A)

G7A01. What is the function of a power supply bleeder resistor?
A. It acts as a fuse for excess voltage.
B. It discharges the filter capacitors when power is removed.
C. It removes shock hazards from the induction coils.
D. It eliminates ground loop current.

G7A02. Which of the following components are used in a power supply filter network?
A. Diodes.
B. Transformers and transducers.
C. Capacitors and inductors.
D. All these choices are correct.

G7A03. Which type of rectifier circuit uses two diodes and a center-tapped transformer?
A. Full-wave.
B. Full-wave bridge.
C. Half-wave.
D. Synchronous.

G7A04. What is characteristic of a half-wave rectifier in a power supply?
A. Only one diode is required.
B. The ripple frequency is twice that of a full-wave rectifier.
C. More current can be drawn from the half-wave rectifier.
D. The output voltage is two times the peak input voltage.

G7A05. What portion of the AC cycle is converted to DC by a half-wave rectifier?
A. 90 degrees.
B. 180 degrees.
C. 270 degrees.
D. 360 degrees.

G7A06. What portion of the AC cycle is converted to DC by a full-wave rectifier?
A. 90 degrees.
B. 180 degrees.
C. 270 degrees.
D. 360 degrees.

G7A07. What is the output waveform of an unfiltered full-wave rectifier connected to a resistive load?
A. A series of DC pulses at twice the frequency of the AC input.
B. A series of DC pulses at the same frequency as the AC input.
C. A sine wave at half the frequency of the AC input.
D. A steady DC voltage.

G7A08. Which of the following is characteristic of a switchmode power supply as compared to a linear power supply?
A. Faster switching time makes higher output voltage possible.
B. Fewer circuit components are required.
C. High-frequency operation allows the use of smaller components.
D. Inherently more stable.

G7A09. Which symbol in figure G7-1 represents a field effect transistor?
A. Symbol 2.
B. Symbol 5.
C. Symbol 1.
D. Symbol 4.

G7A10. Which symbol in figure G7-1 represents a Zener diode?
A. Symbol 4.
B. Symbol 1.
C. Symbol 11.
D. Symbol 5.

G7A11. Which symbol in figure G7-1 represents an NPN junction transistor?
A. Symbol 1.
B. Symbol 2.
C. Symbol 7.
D. Symbol 11.

G7A12. Which symbol in Figure G7-1 represents a solid core transformer?
A. Symbol 4.
B. Symbol 7.
C. Symbol 6.
D. Symbol 1.

G7A13. Which symbol in Figure G7-1 represents a tapped inductor?
A. Symbol 7.
B. Symbol 11.
C. Symbol 6.
D. Symbol 1.

Answer Key

G7A01: B	G7A02: C	G7A03: A	G7A04: A
G7A05: B	G7A06: D	G7A07: A	G7A08: C
G7A09: C	G7A10: D	G7A11: B	G7A12: C
G7A13: A			